

AYUTTHAYA, Thailand

WMO: 484150

Lat: 14.5347N

Lon: 100.7250E

Elev: 10

StdP: 101.21

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.1	19.5	10.9	8.1	21.2	12.8	9.2	22.2	10.5	22.0	9.6	22.1	2.7	50	0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.4	37.5	26.9	36.6	26.9	35.7	26.8	28.6	34.0	28.2	33.4	27.8	33.0	2.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.2	23.1	31.5	26.9	22.5	31.1	26.6	22.1	30.7	92.8	34.4	90.7	33.5	89.1	33.0	31.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
8.1	6.3	5.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			16.0	38.2	2.2	1.5	14.4	39.2	13.1	40.1	11.8	41.0	10.2	42.0
			12.7	29.6	1.7	0.8	11.4	30.2	10.4	30.7	9.4	31.1	8.1	31.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.9	26.5	28.4	29.8	30.8	30.7	29.9	29.3	29.1	28.8	28.0	26.6	
	DBStd	2.03	2.12	1.55	1.84	1.66	1.65	1.20	1.21	0.92	1.03	1.11	2.07	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	6904	513	517	615	624	641	598	599	592	564	583	542	515
	CDD18.3	3863	254	283	357	374	383	348	341	334	314	325	292	257
	CDH23.3	44922	2432	3166	4428	4916	5009	4344	4092	3901	3563	3629	3032	2411
	CDH26.7	20469	1004	1461	2222	2664	2629	2056	1789	1613	1390	1434	1234	972
Wind	WSAvg	1.5	1.9	1.4	1.8	1.6	1.4	1.4	1.4	1.3	0.8	1.1	1.8	2.5
Precipitation	PrecAvg	1140	9	7	38	81	144	122	127	154	273	154	34	5
	PrecMax	1438	49	43	175	216	270	205	331	270	440	414	166	55
	PrecMin	860	0	0	1	7	34	20	50	63	143	37	1	0
	PrecStd	158	12	9	44	55	63	46	55	53	77	73	42	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	35.8	37.5	39.1	38.8	37.4	36.5	34.9	34.6	35.0	34.7	34.9
		MCWB	24.8	25.2	25.8	26.8	27.2	27.1	27.4	27.8	28.1	27.3	26.6	25.4
	2%	DB	33.6	34.7	36.3	37.6	37.4	35.9	34.9	33.9	33.7	34.0	33.9	33.8
		MCWB	24.4	25.4	26.2	26.9	27.0	27.0	26.7	27.3	27.6	26.9	26.3	25.2
	5%	DB	32.7	33.9	35.2	36.5	36.3	34.7	33.7	33.1	32.9	33.2	33.2	32.7
		MCWB	24.1	25.4	26.2	27.1	27.1	27.0	26.7	27.1	27.4	26.9	25.9	24.8
	10%	DB	31.6	33.0	34.3	35.3	35.1	33.5	32.8	32.3	32.1	32.3	32.2	31.5
		MCWB	23.8	25.2	26.2	27.0	27.1	26.9	26.6	26.8	27.2	26.7	25.6	24.1
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.7	27.6	29.1	28.9	28.3	28.3	28.7	29.5	28.2	28.1	27.2
			MCDB	31.9	32.8	33.9	35.6	35.2	33.9	33.8	33.2	33.3	33.0	33.0
2%		WB	26.1	27.0	27.8	28.2	28.3	27.8	27.7	28.0	28.3	27.8	27.5	26.5
		MCDB	31.3	32.7	33.1	34.9	34.2	33.2	32.5	32.6	32.3	32.3	32.3	31.6
5%		WB	25.6	26.5	27.2	27.7	27.8	27.5	27.2	27.5	27.8	27.5	26.9	25.9
		MCDB	30.4	31.9	32.9	34.2	33.4	32.8	32.0	32.0	31.6	31.8	31.5	30.6
10%		WB	25.0	26.0	26.8	27.3	27.4	27.1	26.9	27.1	27.3	27.1	26.3	25.1
		MCDB	29.3	31.2	32.4	33.5	32.9	32.2	31.4	31.4	31.1	31.2	30.4	29.6
Mean Daily Temperature Range	5% DB	MDBR	10.1	9.9	9.3	9.4	8.5	7.7	7.1	6.9	6.6	7.2	8.4	9.6
		MCDBR	11.5	10.6	10.6	10.4	10.1	9.2	8.4	7.8	7.8	8.4	9.4	10.6
		MCWBR	4.1	3.3	2.9	2.8	2.5	2.5	2.5	2.9	3.1	3.0	3.4	3.8
	5% WB	MCDBR	9.7	9.5	9.5	9.6	9.0	8.1	7.6	7.5	7.3	7.7	8.6	9.9
		MCWBR	3.7	3.1	3.1	3.2	3.0	2.8	2.8	3.1	3.2	3.1	3.3	3.8
Clear-Sky Solar Irradiance	taub	0.532	0.617	0.661	0.599	0.511	0.466	0.455	0.457	0.489	0.567	0.509	0.499	
	taud	1.967	1.791	1.724	1.902	2.196	2.324	2.353	2.335	2.234	1.975	2.116	2.082	
	Ebn at Noon	763	717	699	743	799	827	838	846	823	748	780	781	
	Edh at Noon	178	221	242	202	148	129	125	129	143	181	152	155	
All-Sky Solar Radiation	RadAvg	4.90	5.28	5.51	5.94	5.58	5.30	5.00	4.95	4.81	4.82	4.91	4.80	
	RadStd	0.29	0.21	0.37	0.30	0.38	0.28	0.24	0.32	0.25	0.28	0.28	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	+0.22	N/A	N/A	N/A	+0.29	+0.30	N/A	N/A	+102	+102

Nomenclature: See separate page